Literature Note

Save-scumming research offers an explanation of why players reload, while a distinct collection of computational work addresses when to restart a procedure, when to terminate search and how to learn from uncertain outcomes. All of these sources, together, provide both the conceptual background and the methodological tools for studying retry decisions under limited playtime.

Farmer, Kelly and Jones-Rincon (2025), "Save-Scumming: A Novel Exploration of Video Game Play." This conference abstract reports an exploratory study of the motivations underlying save-scumming, framed through self-determination theory. It establishes that the behaviour merits systematic investigation; however, its focus is psychological, and it does not address how a player ought to allocate time across successive events.

Luby, Sinclair and Zuckerman (1993), "Optimal Speedup of Las Vegas Algorithms." This paper shows that restarting a randomised algorithm can reduce the expected time to obtain a solution, and supplies a mathematical basis for comparing restart schedules. Its setting, however, involves the repeated execution of a single procedure, whereas the present project considers multiple distinct game events that compete for the same limited time. Authors' paper.

Lorenz (2016), "Completion Probabilities and Parallel Restart Strategies under an Imposed Deadline." Lorenz examines the probability that a randomised algorithm completes before a deadline under alternative restart strategies, showing that the presence of a deadline alters how a strategy should be evaluated. The present project shares this concern but applies it to the collection of rewards across a sequence of encounters rather than to the completion of a single computational task.

Weitzman (1979), "Optimal Search for the Best Alternative." Weitzman analyses when searching for an additional uncertain reward is worth its cost and derives a rule specifying where to search and when to stop. This is pertinent to weighing the benefit of a further retry against its cost. The model, however, concerns selection among alternatives rather than progression through a fixed sequence of repeatable game events.

DeGroot (2004 reissue), Optimal Statistical Decisions. This book establishes a framework for decision-making under uncertainty based on probabilities, observations, and the consequences of each action. It underpins the Bayesian approach adopted here for cases in which a game's success probability is unknown: a Beta belief represents this uncertainty and is updated as failures are observed.

Russo et al. (2018), "A Tutorial on Thompson Sampling." The tutorial addresses the trade-off between selecting actions that appear rewarding and exploring actions that yield useful information. Its success-or-failure examples are directly relevant to the problem of learning unknown drop probabilities. The present project likewise employs Bayesian updating, but its

decision rule is a simple accept-or-retry policy rather than Thompson sampling. Authors' manuscript.

Badanidiyuru, Kleinberg and Slivkins (2018), "Bandits with Knapsacks." This work investigates learning and reward collection when actions consume limited resources. It supports the treatment of time as a shared resource, in that time spent on one opportunity is unavailable for the remainder. The present project extends this setting with a particular game structure comprising a fixed event order, travel times, and the choice to accept or retry an outcome.

Sargent and Stachurski (2025), Dynamic Programming: Finite States. The book presents methods for selecting actions when current decisions shape future opportunities. It provides the methodological basis for determining whether to retry immediately or to conserve time for later encounters. In the present application, the decision state comprises the current event, the remaining time, and the failures observed to date. Authors' book website.

Research Gap

Existing Research

Save-scumming research mostly explores why players reload and how they perceive the behaviour. Related computational research covers restart strategies, costly search, sequential decision-making and learning under resource constraints. All of these areas offer valuable methods; but, the sources reviewed do not directly address the combined game setting described below.

Unexplored Aspects

The present project centres on several interrelated features:

- **Shared playtime budget:** every consumes time that could otherwise be allocated to another event
- **Fixed event sequence:** players must account for later encounters, including travel time and the possibility of missing an event before the deadline
- **Accept-or-retry decision:** following an unfavourable outcome, the player may retain the result and proceed, or expend additional time in another attempt

- **Known or uncertain odds:** where success probabilities are unknown, a Beta belief is updated after each failure, which may alter whether a further retry is worthwhile

Significance of the Gap

An event may merit a retry when considered in isolation yet become a poor use of time when a more valuable encounter lies ahead. The failed outcomes which are identical may therefore result in different actions depending on the remaining time, the value of later rewards, and the current belief about the probability of success. Analysing each event separately risks overlooking these interdependencies.

Contribution of the Present Project

This project combines these features in a model that allocates time across sequential encounters. A dynamic program then decides whether to accept or retry on the basis of the current event, the remaining time and the failure count. The simulations then compare the resulting policy with simpler strategies, including never retrying, always retrying, fixed retry limits and reward-based rules.

Scope of the Claim

Within the literature reviewed, no study was found that directly combines this full setting. The contribution of the present project is to bring established methods together for this specific problem and to evaluate the conditions under which accounting for future events and uncertain odds improves decision-making.
